

A

## GB1 Deep Mutational Scanning Library



B

## DMS Library Code

```
import poolparty as pp

orf_pool = pp.from_seq(GB1_ORF)
pos = slice(1, 56)

single_pool = orf_pool.mutagenize_orf(          # Single mutant
    num_mutations=1,
    mutation_type='missense_only_first',
    codon_positions=pos,
    prefix='single',
    mode='sequential')

double_pool = orf_pool.mutagenize_orf(        # Pairwise mutant
    num_mutations=2,
    mutation_type='missense_only_first',
    codon_positions=pos,
    prefix='double',
    mode='sequential')

random_pool = orf_pool.mutagenize_orf(        # Random mutant
    mutation_rate=0.1,
    mutation_type='missense_only_first',
    codon_positions=pos,
    prefix='random',
    mode='random',
    num_states=10000)

wt_pool = orf_pool.repeat(times=100, prefix='wt') # WT control

# Combine and generate
dms_pool = pp.stack([single_pool, double_pool, random_pool, wt_pool])
dms_df = dms_pool.generate_library(num_cycles=1, seed=42)
```

C

## Output

&gt;&gt;&gt; dms\_df

| name           | position | wt_codon    | mut_codon   | wt_aa | mut_aa | repeat_idx | seq           |
|----------------|----------|-------------|-------------|-------|--------|------------|---------------|
| single         | 1        | CAG         | TTC         | Q     | F      | None       | ATGTTCTAC...  |
| single         | 1        | CAG         | CTG         | Q     | L      | None       | ATGCTGTAC...  |
| [1,045 rows]   |          |             |             |       |        |            |               |
| double         | 1, 2     | CAG,TAC     | TTC,TTC     | Q,Y   | F,F    | None       | ATGTTCTTC...  |
| double         | 1, 2     | CAG,TAC     | TTC,CTG     | Q,Y   | F,L    | None       | ATGTTCTCTG... |
| [536,085 rows] |          |             |             |       |        |            |               |
| random         | 3        | AAG         | ATC         | K     | I      | None       | ATGCAGTAC...  |
| random         | 1, 4, 6  | CAG,CTG,CTG | TTC,ATG,CAG | Q,L,L | F,M,Q  | None       | ATGTTCTAC...  |
| [10,000 rows]  |          |             |             |       |        |            |               |
| wt             | None     | None        | None        | None  | None   | 0          | ATGCAGTAC...  |
| wt             | None     | None        | None        | None  | None   | 1          | ATGCAGTAC...  |
| [100 rows]     |          |             |             |       |        |            |               |

D

## DAG View

```
>>> dms_pool.print_dag()

dms_pool (pool, n=547,230)
├─ op[5]:stack [mode=sequential, n=4]
│   ├── single_pool (pool, n=1,045)
│   │   └─ op[1]:mutagenize_orf [mode=sequential, n=1,045]
│   │       └─ orf_pool (pool, n=1)
│   │           └─ op[0]:from_seq [mode=fixed, n=1]
│   ├── double_pool (pool, n=536,085)
│   │   └─ op[2]:mutagenize_orf [mode=sequential, n=536,085]
│   │       └─ orf_pool (pool, n=1)
│   │           └─ op[0]:from_seq [mode=fixed, n=1]
│   ├── random_pool (pool, n=10,000)
│   │   └─ op[3]:mutagenize_orf [mode=random, n=10,000]
│   │       └─ orf_pool (pool, n=1)
│   │           └─ op[0]:from_seq [mode=fixed, n=1]
│   └─ wt_pool (pool, n=100)
│       └─ op[4]:repeat [mode=sequential, n=100]
│           └─ orf_pool (pool, n=1)
│               └─ op[0]:from_seq [mode=fixed, n=1]
```